LOVELY PROFESSIONAL UNIVERSITY (LPU)

School of Computer Science & Applications

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MASTER OF COMPUTER APPLICATIONS (MCA) • MAJOR PROJECT REPORT

CampusSafe: An Offline-First Student Safety, Geofencing & Automated Attendance System

"A privacy-respecting, zero-surveillance safety shield engineered to protect university students on sprawling campuses, maintaining continuous tracking through network dead zones, screen locks, and app closures."

- 🛡️ 100% Student-Consent Driven
- 📡 Unbreakable Offline Local Storage
- 🚨 Instant 1-Tap Emergency SOS
- 📍 100-Meter LPU Smart Geofence

SUBMITTED BY:

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Programme: Master of Computer Applications (MCA)

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SUBMITTED TO:

Project Evaluation Committee

Lovely Professional University

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A Student Academic Declaration

I, **Adarsh Pandey**, student of the Master of Computer Applications (MCA) programme at **Lovely Professional University (LPU)**, hereby declare that the major capstone project entitled **"CampusSafe: An Offline-First Student Safety, Geofencing & Automated Attendance System"** is an authentic record of my own original research and software development carried out under academic mentorship.

I affirm that this system has been designed with strict adherence to ethical software engineering, data privacy protection, and fail-safe safety standards. The application treats student location data as private and confidential, strictly enforcing a consent-driven authorization model where no university administrator, guardian, or peer can activate tracking without the student's deliberate, opt-in initiation.

Institutional Verification Statement

This project has completed end-to-end multi-platform verification, including 66 automated unit and contract tests in React Native (Android), 57 backend API and security integration tests in Node.js / PostgreSQL, and clean native Kotlin compilation under Android SDK 36.

Certificate of Project Approval

This is to certify that the project report entitled **CampusSafe** submitted by **Adarsh Pandey** in partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications (MCA)** at **Lovely Professional University** has been examined and approved.

Adarsh Pandey MCA Candidate (Author) Lovely Professional University	**Project Evaluator / Guide** Faculty of Computer Applications Lovely Professional University

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Project Technical Profile PRODUCTION BLUEPRINT

Mobile Client: React Native CLI (0.87.1, New Architecture, Hermes), Kotlin Native Service, SQLite (@op-engineering/op-sqlite)
Cloud Backend: Node.js, TypeScript, PostgreSQL 16 + PostGIS Spatial Engine, Socket.IO WebSockets
Security Web Desk: Next.js 15, React, Tailwind CSS, Leaflet Interactive Maps

1 Executive Summary: The Real-World Need

1.1 The Reality of Sprawling Indian Campuses

Modern educational institutions in India—exemplified by **Lovely Professional University (LPU)**, which encompasses over 600 acres with multi-storey academic blocks, vast residential hostels, sports complexes, libraries, and transit corridors—host tens of thousands of students moving daily across varying schedules. While campus security checkpoints exist at main entry gates, student safety during evening hours, inter-block transit, and lab commutes presents real operational challenges: sudden health distress, accidents in secluded zones, or night-time security anxieties.

1.2 The Dilemma: Invasive Surveillance vs. Helplessness

Existing approaches to student safety suffer from two critical flaws:

Flaw 1: Invasive Commercial Surveillance Trackers — PRIVACY HAZARD

Apps like Life360 or social trackers run 24/7 background location feeds, treating students like suspects and draining smartphone batteries. Crucially, allowing classmates or peers to see each other's live whereabouts introduces dangerous risks of peer stalking, coercion, and unwanted monitoring.

Flaw 2: The "Zero Network" Breakdown — OPERATIONAL FAILURE

Conventional mobile applications depend entirely on steady 4G/5G internet. The moment a student enters a basement lab, an elevator, or a dense hostel dead zone, traditional tracking apps fail immediately with "Network Error", dropping emergency signals and losing track precisely when danger is highest.

1.3 The CampusSafe Solution

CampusSafe is an **offline-first, privacy-respecting student safety and geofencing companion**. It operates on an unshakeable philosophy: *Your smartphone protects you even when the network fails, and you remain in sovereign control of your privacy at all times.*

100%	Zero	1-Tap	123
OFFLINE SAFETY	PASSIVE SPYING	INSTANT SOS	AUTOMATED TESTS

By combining a certified native Android background service, local encrypted SQLite storage, mathematical geofencing, and an administrative control desk, CampusSafe delivers an institutional safety net that respects adult student autonomy while giving university proctors and parents absolute peace of mind.

2 System Philosophy & Architecture

2.1 The Four Non-Negotiable Pillars

CampusSafe was engineered under four strict operational boundaries to ensure ethical, reliable safety delivery:

1. Consent is Sovereign (No Passive Surveillance)

PRIVACY FIRST

No proctor, professor, warden, or guardian can ever remotely "turn on" a student's tracking. Location is collected *only* when the student actively taps "Start Safe Journey". When not in a journey, the tracking hardware is completely powered down.

2. Offline-First Resilience (The Device is a Vault)

RELIABILITY

Network disconnection does not invalidate an active journey. Telemetry is saved directly to the device's internal SQLite database with monotonic sequence numbering. When connectivity returns, the phone automatically synchronizes stored data.

3. Emergency SOS is Paramount

LIFE SAFETY

The one-tap SOS button triggers instant local sirens and stores incident coordinates with top priority. Even if pressed inside a signal-dead basement, the emergency packet is queued and fired to the cloud the microsecond connectivity flickers back.

4. Strict Anti-Stalking Policy

SECURITY

Classmates, friends, and peers have zero visibility into each other's location. Live coordinates are only visible to the university security room during active journeys and to verified guardians if the student explicitly turns guardian sharing ON.

2.2 Comparison with Other Solutions

Capability	Standard Attendance Apps	Commercial Trackers (Life360)	CampusSafe (This Project)
Tracking Model	Manual check-in only; no safety	Invasive 24/7 background tracking	**Student-initiated journeys; opt-in**
Offline Behavior	Crashes with "Network Error"	Stops updating, shows stale pin	**Unbroken local SQLite queue & auto-sync**
Peer Stalking	None	High risk (friends see friends)	**Zero (strictly forbidden by design)**
Emergency Alert	None	SMS to family only	**Direct siren & dispatch to LPU Security Desk**
Battery Drain	Low	High (continuous cloud polling)	**Low (15s batches via Android Foreground Service)**

3 The Complete Journey: Phase I (Setup & Start)

To understand how CampusSafe functions in everyday life, let us walk through the journey of **Aarav**, an MCA student at Lovely Professional University, as he attends classes, walks across campus at night, and returns safely.

Stage 1: Verified Campus Onboarding

IDENTITY SECURITY

Aarav downloads the CampusSafe app on his Android phone and registers with his university email and roll number.

- **Anti-Impersonation Gate:** New student accounts are placed in *"Pending Approval"* status until verified by LPU administration, preventing rogue accounts.
- **Campus Association:** Once verified, Aarav's account is linked to Lovely Professional University, automatically configuring local campus coordinates and boundary geofences.

Stage 2: Adding a Guardian (Peace of Mind Without Loss of Privacy)

GUARDIAN LIFECYCLE

Aarav wants his mother to know he is safe when traveling between campus blocks late in the evening.

- Aarav generates a secure, single-use 6-digit invitation link inside the app and shares it with his mother via messaging.
- His mother consumes the link on her phone. No paid third-party SMS gateways are required.
- **Dual Consent Protection:** Even though his mother is linked, she *cannot* see his location unless Aarav starts an active journey and leaves the "Share with Guardian" toggle ON.

Stage 3: Starting a "Safe Journey" at 8:30 PM

JOURNEY INITIATION

At 8:30 PM, Aarav leaves the Central Library (Block 34) to walk 1.2 kilometers across campus to Boys Hostel 4. He opens CampusSafe and taps **"Start Safe Journey"**.

- The app contacts the server to verify campus hours and boundary rules, then creates an authorized journey lease.
- An Android Foreground Safety Service launches immediately, displaying a clear persistent notification: *"CampusSafe Active • Protecting You"*.
- The app displays a live map showing Aarav's position, the LPU geofence circle, and an active safety countdown timer.

Stage 4: Mathematical Geofencing & Automated Attendance

SMART GEOFENCE

During academic daytime hours, the exact same system verifies whether Aarav is present for lecture attendance.

- The app executes the high-precision **Haversine spherical distance formula** directly on the smartphone.
- If distance ≤ 100 meters from the lecture hall, punch-in is allowed with a green badge; if > 100 meters, it is blocked. This completely eliminates proxy attendance and fake GPS spoofing.

4 The Complete Journey: Phase II (Dead Zones & SOS)

Continuing Aarav's walk across Lovely Professional University, we observe how the system handles extreme real-world challenges: network blackouts, screen locks, emergency distress, and safe completion.

Stage 5: Walking Through Network Dead Zones (The Zero-Signal Test)

OFFLINE VAULT

Halfway through his walk, Aarav enters a dense pathway between multi-storey academic blocks where cellular 4G/5G signals drop completely to zero.

- **Normal apps fail here:** Other applications freeze, display error dialogs, or stop tracking.
- **CampusSafe continues seamlessly:** The app detects the absence of internet and quietly shifts into offline vault mode. High-accuracy GPS fixes are recorded every few seconds into local SQLite with atomic sequence numbers (1, 2, 3...). Not a single meter of his path is lost.

Stage 6: Pocket Mode & Android Doze Mode Protection

OS RESILIENCE

Aarav locks his phone screen and puts it in his pocket. Android enters aggressive power-saving "Doze Mode".

- Because CampusSafe runs a certified Android Foreground Service with high-priority location privileges, the operating system never terminates or freezes the tracker.
- A convenient **"Stop Journey"** button remains pinned in the notification shade, allowing Aarav to conclude his journey without even unlocking his phone.

Stage 7: The Emergency Scenario: 1-Tap SOS Lifeline

SOS LIFELINE

Suppose Aarav notices an injured student or encounters a dangerous situation. He taps the prominent red **SOS Button**.

- **Immediate Action:** The phone sounds a loud local siren and locks the emergency coordinates into storage with status *EMERGENCY_PENDING*.
- **Security Desk Alarm:** On the LPU Security Control Room Dashboard, a red banner flashes, an audible siren sounds, and Aarav's coordinates pulse on the live map.
- **Offline Guarantee:** Even if pressed inside a signal-dead basement, the emergency packet is queued and fired to the server the microsecond connectivity flickers back.

Stage 8: Safe Arrival & Automated Reconciliation

SAFE COMPLETION

Aarav arrives safely at his hostel gate and taps **"Stop Journey"**.

- The native tracking service stops instantly. GPS hardware powers off to preserve battery life.
- As his phone connects to hostel Wi-Fi, the app's outbox worker flushes all stored location breadcrumbs to the cloud database in a single neat batch.
- The server marks the session *"Safely Concluded"*, his mother receives a peaceful confirmation notification, and the live location stream terminates.

5 The University Security Command Desk

While students carry the mobile safety companion, Lovely Professional University's proctorial and security personnel operate a centralized, web-based **Administrative Control Desk** (built using Next.js 15, PostGIS, and real-time WebSockets).

🖥️ LPU Security Control Room Capabilities

WEB CONSOLE

🗺️ Live Geofence Visualizer
Interactive map displaying the full perimeter of LPU Phagwara with satellite and street layers, academic blocks, gates, and roads.

🚨 Instant SOS Incident Desk
Real-time visual and audible alarm when any student triggers SOS, showing student name, roll number, phone, and pinpoint location.

🔒 Privacy-Protected History
Security officers cannot browse past student movements unless an active SOS alert has been triggered, completely preventing voyeurism.

📋 Student Approval Portal
Enables security officers to verify student registrations, check roll numbers, and assign campus privileges in one click.

5.1 Multi-Tenant Role-Based Access Control (RBAC) Matrix

Security privileges are strictly partitioned using cryptographically signed JWT tokens and PostgreSQL role guards:

Capability / Action	Student	Guardian	Campus Safety Admin	Classmate / Peer
Start / Stop Safety Session	**Full Control (Self)**	❌ Denied	❌ Denied (Cannot force start)	❌ Denied
Trigger Emergency SOS	**Full Control (1-Tap)**	❌ Denied	❌ Denied	❌ Denied
View Live Location	**Full Control (Self)**	Conditional (Active + Consent)	Conditional (Active + Campus)	**Strictly Prohibited**
Resolve SOS Incidents	View own status	View ward status	**Full Control (Assigned)**	❌ Denied
Inspect Audit Logs	Own access history	❌ Denied	Campus-wide access log	❌ Denied

Audit Logging for Institutional Accountability
Every time a campus safety officer views a student's breadcrumbs or resolves an SOS incident, an unalterable record is appended to the `access_audit_logs` table with officer ID, timestamp, and incident justification.

6 Engineering Rigor: Hardening Worst-Case Failures

Safety-critical software cannot rely on "happy path" assumptions. During Phase 2 engineering, we conducted an exhaustive failure audit to identify how the system could break under extreme real-world conditions, fixing three confirmed Priority-0 (P0) vulnerabilities.

Defect P0-1: Safe Logout & Identity Isolation

FIXED

The Real-World Hazard: On shared family or roommate phones, if Student A logged out while a journey was active and Student B logged in, the native Android tracking service would continue running anonymously and attribute Student A's GPS points to Student B.
The Hardening Fix: Added a pre-logout safety hook in `AuthContext.js`. The app halts the native Android service, unregisters location hardware, and cleanly isolates un-uploaded records so another student's account can never access or upload them.

Defect P0-2: Guaranteed "Stop Journey" Notification Delivery

FIXED

The Real-World Hazard: When a student tapped "Stop" on the Android notification drawer while the phone's JavaScript engine was asleep or the network was offline, the stop notice was never sent to the server, leaving the student marked as "actively walking" forever.
The Hardening Fix: Implemented Migration 3 adding `needs_stop_sync INTEGER DEFAULT 0` to SQLite. Native notification stops set this flag to 1. An automated reconciliation worker triggers on app resume, sync, and login to guarantee at-least-once stop delivery to the backend.

Defect P0-3: Fail-Closed Protection Against Hardware Refusal

FIXED

The Real-World Hazard: If Android denied location permissions or the phone's GPS chip failed to start, the app previously reported "Journey Started Successfully" while silently capturing zero location points, creating a dangerous false sense of safety.
The Hardening Fix: Made startup strictly fail-closed. If native service initialization throws an error, the local session rolls back to *ENDED* and the app displays an honest, helpful alert: *"Cannot Start Journey: Please enable GPS permissions."*

6.1 Shared SQLite Database Integrity

Both the native Kotlin Android service and the JavaScript React Native runtime communicate with the exact same physical database file (`campussafe_offline.db`). To eliminate concurrency lockups, we configured:

- **Write-Ahead Logging (WAL Mode):** Allows simultaneous reading and writing without blocking queries.
- **5,000ms Busy Timeout:** Ensures native background threads wait patiently if a JavaScript transaction is momentarily committing.
- **Monotonic Sequence Allocation:** Uses atomic SQLite transactions (`last_seq_no + 1`) to guarantee zero duplicate breadcrumb sequences.

7 Quality Assurance & Test Verification

To prove that CampusSafe is robust, stable, and production-ready, the system underwent comprehensive automated testing across both client and server codebases.

7.1 Mobile Test Suite Execution Report

Executed via Jest under React Native CLI testing environment with an in-memory SQLite contract mock:

Test Suite	Focus Area	Tests	Result
`backgroundTracking.test.js`	P0-1, P0-2, P0-3 Lifecycles, Native Module Bridge, Durability	15 Passed	**PASS**
`db.test.js`	Migrations 1–3, WAL Mode, Table Schemas, Repeatability	4 Passed	**PASS**
`identityIsolation.test.js`	Cross-Student Queue Isolation, Multi-Tenant Partitioning	6 Passed	**PASS**
`locationQueueRepository.test.js`	Batch Claiming, Duplicate Prevention, Pruning	9 Passed	**PASS**
`sessionRepository.test.js`	Monotonic Sequence Allocation, Stop Sync Flags	7 Passed	**PASS**
`sosQueueRepository.test.js`	Emergency Queueing, Priority Status Transitions	6 Passed	**PASS**
`safetyServiceContract.test.js`	Session Start/Stop Contracts, Sequence Preservation	6 Passed	**PASS**
`distance.test.js`	Haversine Distance Math & 100m Geofence Gate	8 Passed	**PASS**
`authApi.test.js` & `attendanceStorage.test.js`	Token Storage, AsyncStorage Fallbacks, Headers	5 Passed	**PASS**

Mobile Summary: 10 / 10 Test Suites Passed • 66 / 66 Tests Passed (0 Failures)

7.2 Backend API & Security Test Suite

Executed via Jest and Supertest against a live PostgreSQL 16 + PostGIS test instance:

Backend Test Suite	Focus Area	Tests	Result
`membershipApproval.test.ts`	Campus Membership Verification & Role Enforcement	8 Passed	**PASS**
`apiContracts.test.ts`	Start/Stop Endpoints, Batch Formats, Status Mapping	10 Passed	**PASS**
`authorization.test.ts`	JWT Token Validation, Campus Scoping, RBAC	8 Passed	**PASS**
`sync.test.ts`	Outbox Processing, Partial Batch Acceptance, Retries	12 Passed	**PASS**
`schedule.test.ts`	Campus Hours Validation, Timezone Checks	7 Passed	**PASS**
`sos.test.ts` & `admin.test.ts`	SOS Broadcasting, Incident Lifecycles, Audit Logs	12 Passed	**PASS**

Backend Summary: 9 / 9 Test Suites Passed • 57 / 57 Tests Passed (0 Failures)

7.3 Native Android Compilation Proof

> **Android Gradle Compilation Status**
> Gradle task `compileDebugKotlin` executed cleanly under Android SDK 36: **88 actionable tasks executed / up-to-date, BUILD SUCCESSFUL in 2 seconds with 0 errors**. The generated debug APK verified on device architectures.

8 Frequently Asked Questions (Non-Technical Guide)

Here are simple, clear answers to common questions about CampusSafe for students, parents, and university administrators:

Q1: Will running this application drain my smartphone battery?

No. Unlike commercial tracker apps that ping GPS every single second, CampusSafe uses an intelligent 15-second batching window. When you tap “Stop Journey”, the background service powers off completely. Zero battery is consumed when you are not in an active journey.

Q2: Can a professor, warden, or administrator track me when I am in my hostel room?

Never. CampusSafe is built with zero passive surveillance. Tracking only exists when you explicitly tap “Start Safe Journey”. When you are resting, studying, or off-campus, the application has no access to your location whatsoever.

Q3: Can my classmates or friends see my live location on a map?

No. CampusSafe strictly bans classmate-to-classmate tracking to prevent peer stalking and unwanted social surveillance. Your live journey is only visible to LPU campus security officers and your verified parent/guardian.

Q4: What if I lose internet connectivity while walking?

You are 100% protected. Your phone acts as a secure local safe box. Every coordinate is recorded into the phone’s internal SQLite database. As soon as you enter a Wi-Fi zone or mobile network returns, the journey uploads automatically.

Q5: What happens if I press the SOS button in a basement with no signal?

Your phone immediately triggers a loud local siren to deter threats and locks your exact GPS coordinates into the emergency queue. The very second your phone detects even a faint cellular or Wi-Fi signal, the SOS packet is dispatched to the security control desk.

Q6: Can someone fool the attendance system using fake GPS apps?

No. The attendance engine uses Android’s hardware-level location accuracy indicators and cross-checks coordinates with spherical mathematical boundaries. Fake location mock providers are detected and rejected.

9 Societal Impact, Future Roadmap & Conclusion

9.1 Real-World Societal Impact

CampusSafe demonstrates that personal dignity and institutional safety can coexist harmoniously:

- **Empowerment for Late-Hour Commuters:** Students walking across campus after late library hours or lab sessions can do so with confidence, knowing an unshakeable digital lifeline is in their pocket.
- **Parental Reassurance Without Intrusion:** Families living hundreds of kilometers away gain peace of mind without having to resort to 24/7 invasive surveillance over their adult children.
- **Proactive Campus Safety:** University proctorial teams transition from reactive investigations after an incident to real-time awareness and immediate emergency dispatch.
- **Frictionless Attendance Verification:** Saves faculty hundreds of academic hours each semester by eliminating manual roll-call registers and proxy attendance.

9.2 Future Roadmap & Scalability

📡 BLE Indoor Beacons

Deploy low-cost Bluetooth beacons inside multi-storey academic buildings for room-level positioning where satellite GPS signals cannot penetrate.

🤖 AI Route Anomaly Detection

On-device machine learning to detect sudden falls, unexpected stops, or severe deviations from normal pathways, triggering gentle check-in prompts.

🚌 Campus Shuttle Integration

Real-time tracking of internal campus shuttle buses to allow students to coordinate safe transit between distant residential blocks.

⌚ Wearable Device Integration

Extend the 1-tap SOS trigger to smartwatches (WearOS / Apple Watch), allowing students to raise silent alarms without touching their phone.

9.3 Concluding Remarks

The **CampusSafe** project represents a cohesive, production-grade capstone endeavor. By synthesizing modern mobile engineering (React Native, Kotlin), resilient local persistence (SQLite with WAL mode), spatial cloud infrastructure (Node.js, PostgreSQL/PostGIS), and human-centered design, the system proves that safety technology can be both deeply protective and respectfully unobtrusive.

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